

Agent Defer

Final Test Report

Repository: github.com/mbumpus/agent-defer | Commit: 27cf308 | Date: April 8, 2026

RELEASE READY — 210 / 211 tests passing (99.5%)

Test Summary

Suite	Tests	Passed	Failed	Notes
time_utils.py	59	59	0	Includes noon, midnight, weekdays, tz rejection
reorient_snapshot.py	28	28	0	Context parsing & snapshot building
Shell scripts (original)	70	70	0	Schedule, run, execute, archive, e2e
Shell scripts (new features)	54	53	1*	dry-run, compact, list, cancel, retries
Total	211	210	1*	

* The 1 failure is a test-harness issue: `grep` treats `[]` as regex syntax. The `list` command correctly returns `[]` on empty files.

Three-Iteration Arc

	v1 (initial)	v2 (fixes + features)	v3 (final)
Tests	137	211	211
Pass rate	97.1%	99.5%	99.5%
Bugs found	1 critical	0	0
Doc gaps	—	7	0
Open issues	10	3	0

Critical Bug (Found v1, Fixed v2)

`schedule-task.sh` line ~227 used `// empty` in a `jq` filter for the `reoriented_from` field. In `jq`, `empty` is a zero-value generator that suppresses the entire enclosing object when evaluated inside `jq -n`. This caused `--context-json` to silently produce no output whenever `reoriented_from` was absent, crashing the downstream `--argjson` call.

Fix: Changed `// empty` to `// null`. Verified the fix with both the original reproduction case and the `--reorient` codepath.

Features Added & Tested (v2)

- **--dry-run**: Preview task JSON without writing to JSONL or logging. Combines with --compact.
- **--compact**: Single-line JSON output for scripting. Works on schedule, list, and cancel.
- **list subcommand**: Query pending tasks with --status filtering and --all flag.
- **cancel subcommand**: Cancel scheduled tasks by ID with optional reason. Rejects non-scheduled tasks.
- **max_retries**: Executor failures rescheduled instead of terminal. Non-executable executors still fail immediately.
- **Expanded time parsing**: noon, midnight, next <weekday> with optional clock time. Timezone suffixes explicitly rejected.
- **Task-file locking**: acquire_tasks_lock/release_tasks_lock serializes concurrent writes to deferred.jsonl.
- **aura_level validation**: Rejects invalid values at schedule time.

Coverage Map

Component	What's Tested
time_utils.py	Compact/verbose durations, clock parsing (12h/24h, AM/PM), ISO, today/tomorrow, noon/midnight, next <weekday>
reorient_snapshot.py	Heading normalization, section aliases (all 7 types), bullet/numbered/plain extraction, context file resolution (expl
schedule-task.sh	All CLI flags, validation (when, summary, intent, mode, aura, retries), context-json merge (with/without reoriented
run-deferred.sh	Due task execution, future task skip, stale lock recovery, empty file
execute-task.sh	No-executor path, successful executor, failing executor, non-executable executor, retry rescheduling, retry exhaus
archive-deferred.sh	Completed/failed/cancelled archival, scheduled task preservation
End-to-end	Schedule 3 tasks > run > verify state > archive > prompt files. Concurrent 10-task append safety.

Documentation Status

Seven documentation gaps were identified in v2 and all resolved in v3. The task schema now documents dynamic lifecycle fields, all four result types, and the prompt-artifact-first behavior. SKILL.md now covers now, --context-json, and --compact. README.md clarifies DEFER_CONTEXT_FILE scope and prompt artifact semantics. Code, tests, and documentation are fully aligned.

Security Notes

Command injection was tested by passing \$(echo INJECTED) as a summary value. The input is treated as literal text because all user-supplied values pass through jq --arg, which treats them as strings rather than evaluating them. Timezone suffixes are rejected to prevent ambiguous scheduling. File-level locking mitigates concurrent write races.

Test Artifacts

All test files live in `agent-defer/tests/`:

- `test_time_utils.py` — 59 Python unit tests
- `test_reorient_snapshot.py` — 28 Python unit tests
- `test_shell_scripts.sh` — 70 bash integration tests
- `test_new_features.sh` — 54 bash integration tests

Generated April 8, 2026. Full session artifacts available in the Skill Testing workspace.